

Solar-Powered Environmental Sensing Node

Hardware design take-home, Rev A architecture and schematic design

Candidate: Jyotiraditya Ingawale | Design time box: approximately 6 hours

1. Overview

This node runs an environmental sensor and an off-board UART radio from a nominal 12 V solar panel and a 2S1P LiFePO4 pack, unattended, for at least 12 months in rural southern Africa. Ambient temperature spans -5 C to +45 C and a sealed enclosure in direct sun may reach +70 C internally.

The architecture rests on three decisions. A BQ25798 buck-boost charger provides MPPT, a fully programmable LiFePO4 charge profile, and NVDC (Narrow-Voltage DC) power-path management, so the electronics keep running from solar even while battery charging is inhibited, for example when the pack is too hot. A BQ76905 monitor gives the battery an independent protection layer that does not depend on correct charger configuration or correct firmware. Finally, charging is disabled by default in hardware and enabled only after the MCU has programmed the charger and read back a verified profile, so loss of MCU control fails safe.

The headline numbers are a 10 W panel, a 300 mA charge at 7.30 V, a 5-minute reporting interval, a design load near 1.31 Wh/day dominated by the radio's 10 mA idle current, and roughly one week of battery-only autonomy. A routed PCB was outside the time box; Section 10 gives the floorplan and the routing constraints that matter most.

2. System architecture

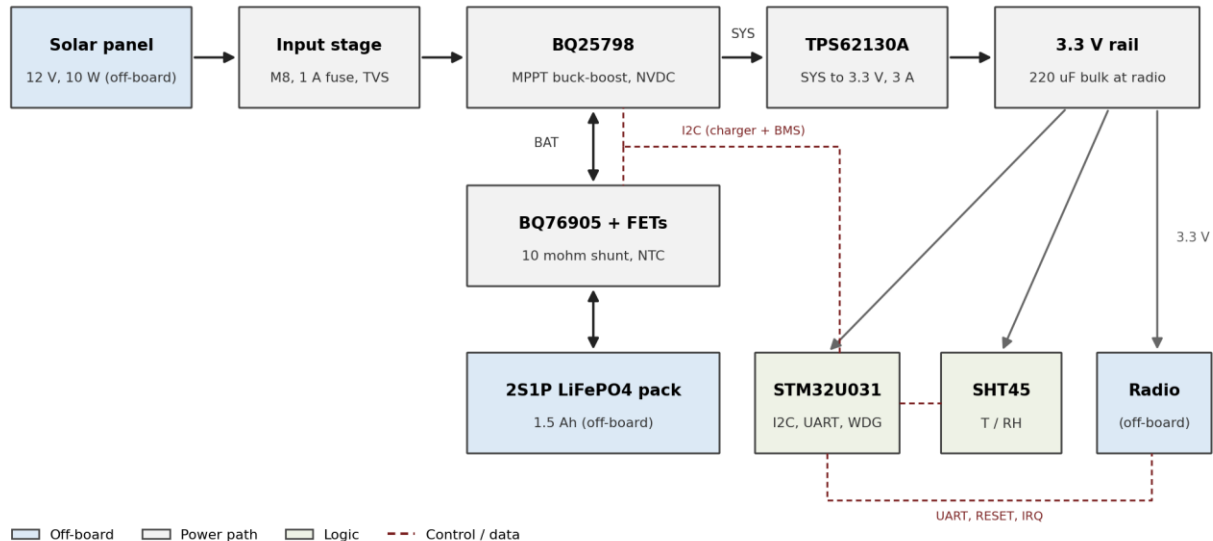


Figure 1. Power path (solid) and control paths (dashed). The pack, panel, and radio are off-board.

Power flows from the panel through the fused input stage into the BQ25798, which serves the SYS rail and manages battery charge and discharge through the BQ76905 protection domain. The TPS62130A generates 3.3 V for the MCU, sensor, and radio connector. The STM32U031 configures the charger and reads BMS telemetry through a domain-aware I2C interface, samples the SHT45, and talks to the radio over UART with a dedicated hardware reset line and a status interrupt. The NVDC path matters at high enclosure temperature: when pack

temperature forces a charge inhibit, the node continues operating on whatever solar energy is available instead of draining the battery.

3. Solar input and panel sizing

I selected a 10 W monocrystalline 12 V-class panel. The electrical need is modest, about 1.31 Wh/day at the design load, so sizing is driven by margin rather than steady-state demand: elevated panel temperature, dust accumulation over a maintenance-free year, imperfect tilt, seasonal rain, aging, and recovery after several low-harvest days. A 5 W panel would likely suffice under nominal conditions. I rejected a 20 W panel because the pack absorbs only about 2.1 W at the 300 mA charge rate, so the extra area buys wind loading, cost, and mounting complexity without improving normal operation.

The panel enters through a keyed 2-pin M8 bulkhead connector, which separates the weather-sealing problem from PCB assembly, followed by a 1 A one-time fuse and a transient-protection footprint. The panel's short-circuit current is only about 0.6 A, so 1 A gives a sensible margin. A one-time fuse is preferred over a resettable PTC because PTC hold current derates steeply toward +70 C, while a fixed fuse keeps a predictable rating. Reverse-polarity protection relies on the keyed connector, a controlled harness, and clear labeling rather than a series MOSFET, since the BQ25798 input structure makes conventional PFET protection less straightforward than on a simple regulator. The TVS is deliberately left as a footprint pending clamp analysis: a 36-cell panel's open-circuit voltage approaches 24 V on a cold morning, right at the top of the charger's 3.6 to 24 V operating range, so the clamp must stay off at cold VOC while still holding transients under the 30 V absolute maximum. Direct lightning protection is outside Rev A scope.

4. Charging and power path

I chose the BQ25798 over a fixed-profile standalone solar charger for four reasons. It supports 1 to 4 cells, including LiFePO₄ chemistry, with charge voltage and current programmed over I²C, so a future change of cell vendor or approved charge rate is a firmware edit rather than a power-stage redesign. Its four-switch buck-boost topology tolerates panel voltages above and below the pack voltage. It implements solar MPPT and NVDC power-path management. Its integrated ADC provides input voltage, charge current, and fault telemetry that feeds the remote diagnostics in Section 9. The cost is configuration complexity in a part whose autonomous 2S defaults target conventional lithium-ion rather than LiFePO₄, which motivates the initialization scheme below.

The Rev A profile is VREG = 7.30 V and ICHG = 300 mA, the reference cell's standard charge rate. The low rate trades charge speed for cell life and thermal margin, both of which matter more than speed when harvest exceeds demand by an order of magnitude.

Safe initialization is a hardware property, not a firmware promise. The active-low CE pin is pulled high from the charger's REGN rail, so charging is disabled whenever the MCU is absent, resetting, or unconvinced. An MCU-driven N-channel MOSFET pulls CE low only after the STM32 has booted, established I²C contact, programmed the LiFePO₄ profile, read the critical registers back, and confirmed the values. The firmware enable and the hardware CE path form two independent conditions that must both agree before the pack sees charge current.

The converter runs at 750 kHz with a 2.2 μ H inductor. I chose the lower-frequency option over the more compact high-frequency mode because board area is available and reduced switching loss helps thermal margin in a +70 C enclosure. Input current limiting is layered: the firmware IINDPM target of roughly 0.5 A sits under the panel's 0.6 A capability, and a hardware ILIM ceiling near 0.8 A guarantees that firmware is never the only mechanism limiting source demand.

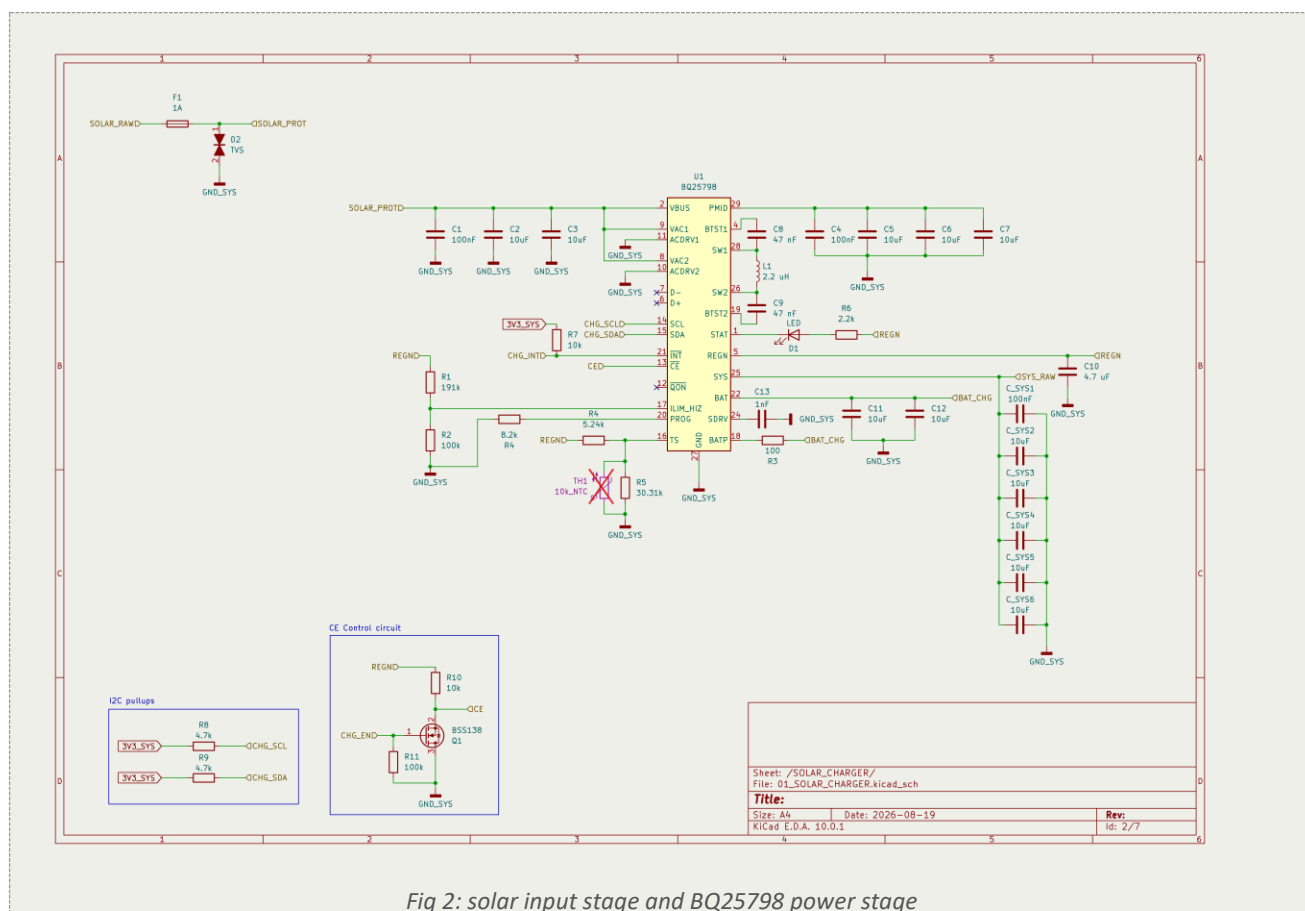


Fig 2: solar input stage and BQ25798 power stage

5. Battery protection and temperature policy

The BQ76905 provides the battery's independent safety boundary: per-cell voltage measurement, low-side NFET drivers for charge and discharge paths, overcurrent and short-circuit protection, overvoltage and undervoltage protection, an external NTC input, open-wire detection, coulomb counting, and host-controlled balancing, at a typical operating current near 140 uA. The charger manages normal energy flow; the BMS guarantees that a firmware bug or a misconfigured charger is never the last line of defense for the cells. For the 2S configuration, unused voltage channels collapse per the manufacturer's guidance, and the active cell inputs receive 10 ohm / 0.22 uF RC filters.

Pack current is measured across a 10 milliohm shunt: 10 mV at 1 A and 45 mV at 4.5 A, comfortably inside the roughly +/-200 mV measurement range, with far better resolution than the 1 milliohm shunts common in high-current reference designs. Dissipation at 4.5 A is about 0.20 W, so a 1 W resistor carries good margin. The SRP/SRN traces must be Kelvin-routed from the shunt terminals. I kept the protection MOSFETs on the TI reference topology rather than optimizing them for cost, which keeps prototype risk low.

Battery temperature is treated separately from board temperature through a dedicated 10 kohm NTC bonded to the pack:

Pack temperature	Intended behavior
Below -10 C	Battery use inhibited
-10 to 0 C	Discharge permitted, charging inhibited
0 to 45 C	Normal charge and discharge
45 to 60 C	Charging inhibited, discharge may continue
Above 60 C	Battery isolated, critical thermal fault

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Fig 3: BQ76905 cell-monitoring and current-sense implementation;
The protection FET stage follows the TI reference topology and is reserved for detailed implementation.

A TPS62130A buck generates 3.3 V from SYS. The radio defines the peak: 1 A transmit plus MCU and miscellaneous loads gives a design peak near 1.15 A, or about 3.8 W. A mature 3 A part with fast transient response leaves generous margin, and the circuit follows the manufacturer's reference implementation.

Pin	Function	Pin	Function
1	3V3	5	UART TX
2	3V3	6	UART RX
3	GND	7	RESET
4	GND	8	IRQ / STATUS

The connector receives local capacitance of roughly 220 uF bulk, 22 uF ceramic, and 100 nF ceramic to buffer burst current against harness impedance. UART lines carry modest series damping with ESD provision near the connector. No load switch is included: the assignment states the radio has no sleep mode, so Rev A powers it continuously and accepts that its idle current dominates the energy budget. The MCU-controlled RESET line provides recovery if the radio firmware locks up.

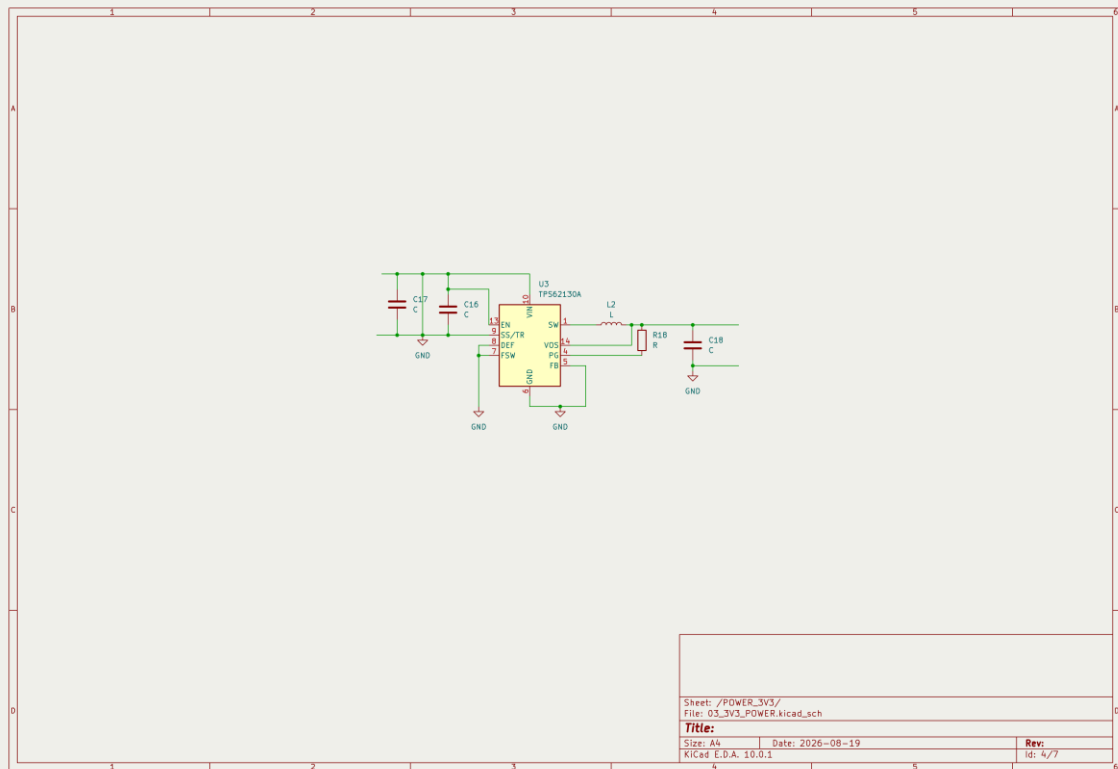


Fig 4: TPS62130A regulator

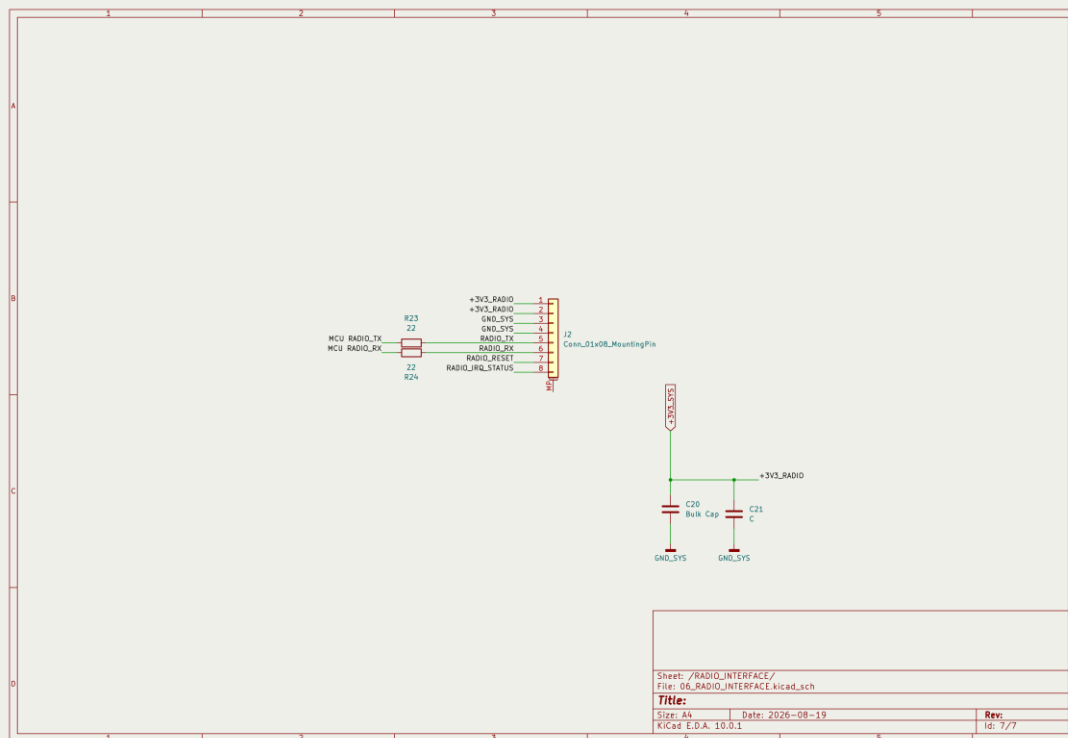


Fig 5: Radio Interface

7. MCU, sensor, and firmware behavior

The STM32U031K8 matches the actual workload, which is charger configuration, BMS telemetry, sensor acquisition, UART communication, fault handling, and watchdog supervision. It offers low-power operation in a compact implementation without paying for DSP, USB, Ethernet, or external memory the application will never use. Interfaces: I2C to the charger and SHT45, a domain-aware interface to the BQ76905, UART plus reset and status GPIO to the radio, the charger interrupt and hardware charge-enable, SWD for debug, and two status LEDs for commissioning that stay off in normal operation.

The independent watchdog is refreshed only when the charger manager, scheduler, and communication tasks all report healthy, never unconditionally. The MCU records the reset cause at startup, and reset statistics ride along in telemetry so a field team can distinguish a brownout from a firmware lockup without a site visit.

The environmental sensor is the SHT45-AD1F: temperature and relative humidity over I2C, 1.08 to 3.6 V supply, typical average current near 0.4 uA, typical accuracy of +/-1 %RH and +/-0.1 C, with a factory PTFE membrane for particle protection. The electrical interface is trivial; the real problem is obtaining a representative ambient reading. The sensor sits at the outer PCB edge behind thermal isolation slots, as far as practical from the charger, buck, inductors, and radio power path, aligned with a shaded, downward-facing ePTFE enclosure vent. I considered a four-wire daughterboard behind the vent and deferred it: it improves thermal isolation but adds a connector, a harness, and a field failure point. If prototype thermal testing shows unacceptable bias against a reference probe, the daughterboard is the fallback.

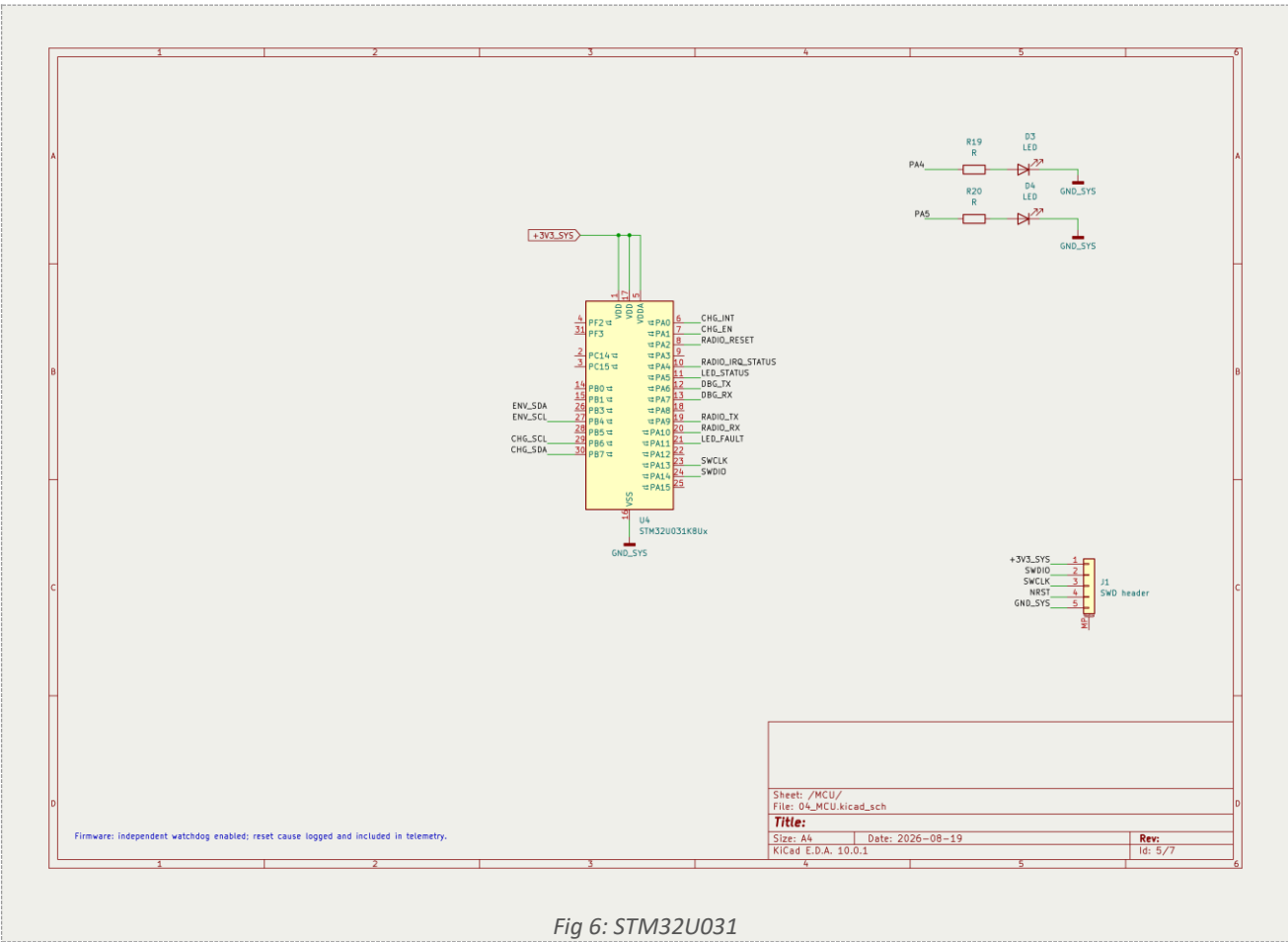


Fig 6: STM32U031

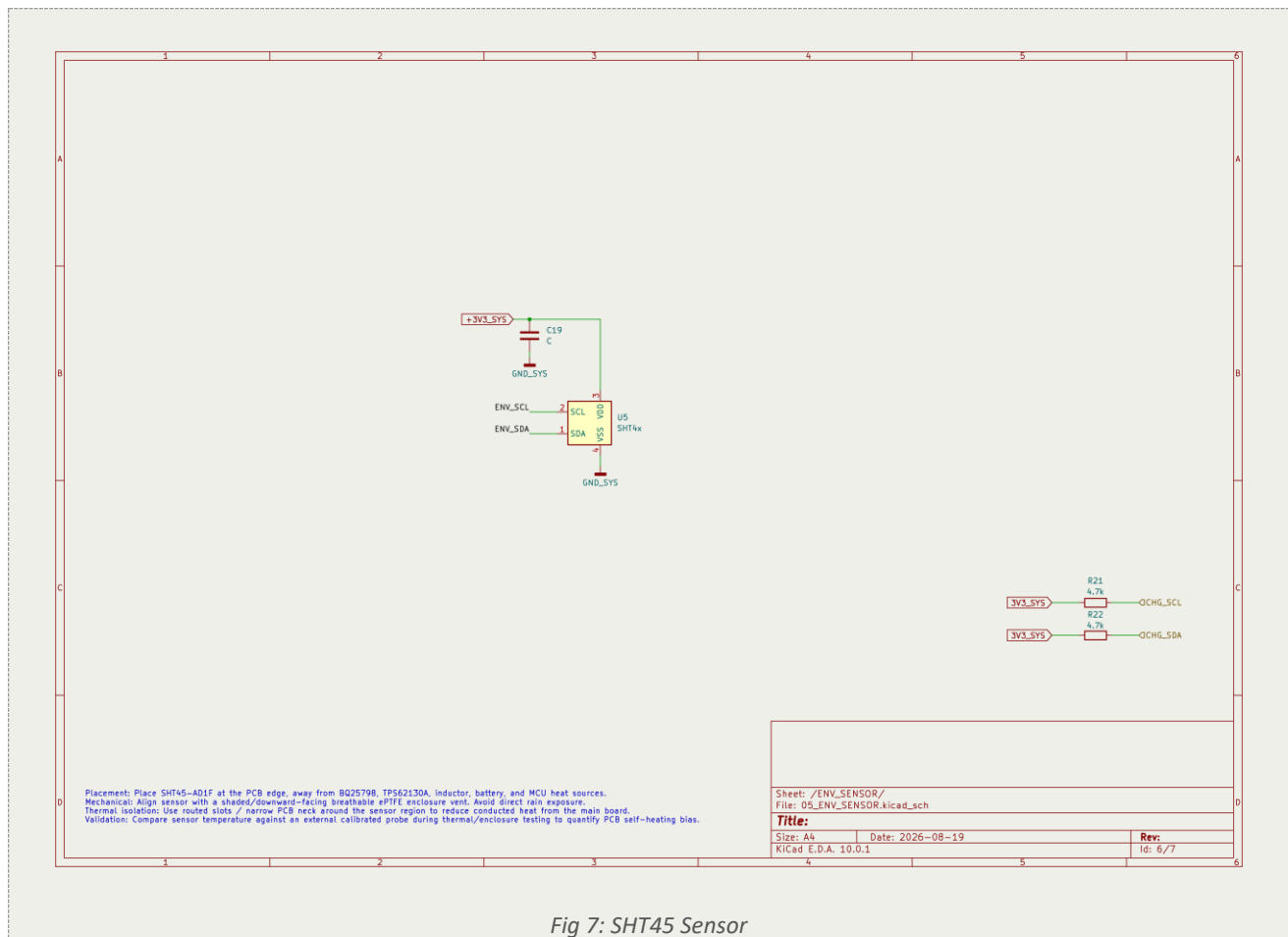


Fig 7: SHT45 Sensor

Firmware sequencing:

boot:

```
init clocks, GPIO, I2C, UART, RTC, independent watchdog
read and log reset cause
CE stays high: charging disabled by hardware default
program BQ25798: VREG 7.30 V, ICHG 300 mA, IINDPM 0.5 A, MPPT on
read back critical registers
if profile verified: assert CHG_EN, pulling CE low
else: leave charging disabled, raise fault flag
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every 5 min (RTC scheduled):

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sample SHT45; mark measurement invalid on I2C failure
read charger ADC: VBUS, VBAT, ICHG, status, faults
read BQ76905: cell voltages, pack current, pack temp, protection state
apply temperature policy with hysteresis (charge window 0 to 45 C)
build telemetry frame: measurements, counters, reset cause, faults
transmit over UART; on timeout retry, then pulse RADIO_RESET, retry once
refresh watchdog only if charger, scheduler, comms tasks are healthy
enter low-power stop until next RTC tick
```

8. Energy budget and autonomy

With transmit current $I_{TX} = 1$ A for 0.5 s and idle current 10 mA, average radio current over interval T is $I_{avg} = I_{idle} + (I_{TX} - I_{idle})(0.5/T)$:

Report interval	Average current	Average power at 3.3 V	Energy per day
1 min	18.25 mA	60.2 mW	1.445 Wh
5 min	11.65 mA	38.4 mW	0.923 Wh
15 min	10.55 mA	34.8 mW	0.836 Wh
60 min	10.14 mA	33.5 mW	0.803 Wh

I selected a 5-minute interval. It cuts radio energy by 36% relative to 1-minute reporting, while stretching from 5 to 15 minutes saves only about 9% more, because the specified 10 mA idle current dominates the budget once the radio is continuously powered. The interval affects latency, energy, and channel airtime; it does not change range for a fixed modulation.

Subsystem	Estimated energy per day
Radio at 5 min interval	0.923 Wh
STM32U031	~0.001 Wh
SHT45	<0.001 Wh
BQ76905	~0.02 Wh
BQ25798 quiescent and control	~0.003 Wh
Conversion and miscellaneous losses	~0.10 Wh
Nominal total	~1.05 Wh

A 25% engineering allowance covers conversion uncertainty, temperature, aging, tolerance, wiring loss, and unmodeled loads, giving a design load of about 1.31 Wh/day.

The pack stores $6.4 \text{ V} \times 1.5 \text{ Ah} = 9.6 \text{ Wh}$ nominal. Applying a conservative 80% usable-energy factor gives 7.68 Wh, which supports about 7.3 days at the nominal load and 5.9 days at the design load. Three fully sunless days consume 3.93 Wh, roughly half the usable store, so the node rides out multi-day weather with margin. I considered an energy-adaptive reporting mode that slows the cadence at low state of charge and deliberately excluded it from Rev A so the baseline behavior and the quoted budget stay deterministic.

Harvest is estimated as $E = P \times H \times \eta$ with a lumped η of 60% covering panel temperature, orientation error, dust, wiring, charger loss, and imperfect MPPT:

Condition	Equivalent sun hours	Harvest estimate
Good	5.5 h	33 Wh/day
Conservative	3 h	18 Wh/day
Poor	1 h	6 Wh/day

Even the poor scenario exceeds the 1.31 Wh/day design load several times over. The practical limits are therefore consecutive no-sun days, battery temperature restrictions on charging, severe panel contamination, and installation geometry rather than steady-state panel power. Site-specific solar modeling should follow once the deployment location is known.

9. Thermal strategy, fault handling, and diagnostics

The BQ25798 is rated to +85 C, which leaves limited headroom above a +70 C enclosure, so self-heating is managed at the source: the lower 750 kHz switching option, compact power loops, copper spreading on the power layer, thermal vias under exposed pads, physical separation between power components and the SHT45, and charge inhibit driven by pack temperature. I rejected active cooling because fans and Peltier devices add power draw, ingress paths, moving parts, and maintenance to a product that must survive a year untouched. The enclosure should be UV-resistant and light-colored, with sealed bulkheads, no upward-facing water entry, a sheltered sensor vent, and a panel angle that promotes drainage and some self-cleaning.

Fault management is layered. The charger handles normal energy flow. The BQ76905 autonomously protects the cells regardless of what the charger or firmware believes. The MCU supervises configuration, diagnostics, and recovery under a conditionally serviced watchdog. Physical protection closes the stack with the input fuse, protection FETs, controlled connectors, and conductor sizing. The governing principle is that a fault degrades functionality before it compromises battery safety.

Fault	Intended response
Solar absent or input fuse open	Continue from battery while SOC allows
Battery too hot to charge	Stop charging, run from solar where available
Charger profile invalid or comms lost	CE stays high, charging disabled, fault reported
Cell OV, UV, overcurrent, short	BQ76905 opens the protection FET
NTC open or short	Inhibit charging, report fault
Sensor comms failure	Continue operating, mark measurement invalid
Radio unresponsive	Hardware reset and retry
MCU software lockup	Watchdog reset, cause logged

Telemetry covers solar and charger state (input voltage, charge current, MPPT and fault status), battery state (per-cell voltages, pack current, pack temperature, protection and balancing status), system health (uptime, boot count, reset cause, watchdog count, transmit count), and the environmental measurements themselves. This lets a field team distinguish a failed panel from a depleted battery, an overheated pack, a resetting MCU, or a dead radio before dispatching anyone.

10. PCB floorplan and layout priorities

The board is constrained to 100 mm x 50 mm on a four-layer 1 oz process: L1 components and priority routing, L2 primarily solid GND_SYS plane, with intentional battery-domain separation around the low-side protection boundary, L3 power distribution and slower signals, L4 signals and ground fill. A fully routed layout was a deliberate scope cut; the floorplan and constraints below are the contract any routing pass must honor.

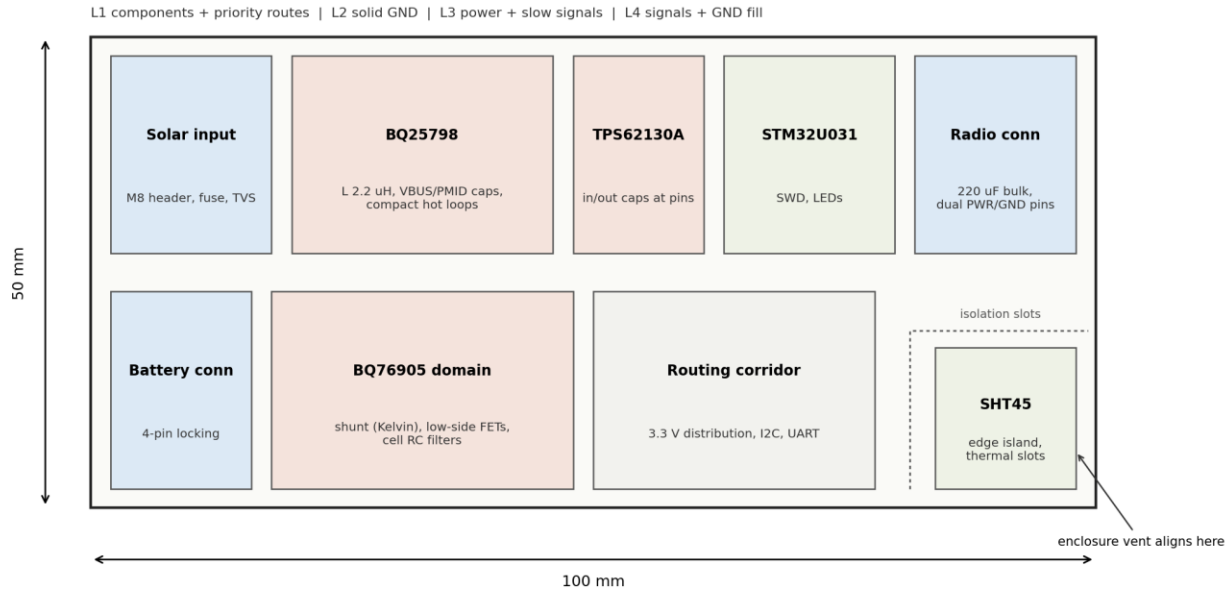


Figure 8. Floorplan to scale. Power flows left to right along the top edge; the battery protection ground domain occupies the lower left; the SHT45 sits on an isolated island in the corner farthest from the switching converters.

Charger: minimize the switching current loops, place VBUS and PMID capacitors against the IC, keep SW1/SW2 copper small, put bootstrap capacitors at their pins, and stitch thermal vias under the exposed pad. Buck: follow the reference layout with input and output capacitors tight to the part. BMS: Kelvin-route SRP/SRN from the shunt terminals, keep cell RC filters at the monitor, preserve the low-side protection ground-domain structure, and keep the FET and shunt current path compact. Radio: bulk capacitance beside the connector and a low-resistance 3.3 V path. Sensor: outer edge, maximum distance from power components, thermal slots, vent alignment.

11. Assumptions

Assumption	Design impact
Panel is nominal 12 V, ~10 W	Input and charger design centered on 12 V-class PV
Pack is 2S1P LiFePO ₄ , ~1.5 Ah, no integrated BMS	6.4 V nominal, 9.6 Wh; board provides all protection
Cell midpoint and pack NTC are accessible	Per-cell measurement and real pack-temperature control
Charge current target is 300 mA	Cell life and thermal margin over charge speed
Radio stays continuously powered	10 mA idle current dominates the energy budget
Reporting interval is 5 minutes	Balance of latency and energy
Enclosure may reach ~70 C internally	Derating, charge inhibit, sensor isolation required
Prototype is not certified	Standards treated as design inputs, not claims

Standards are treated as design inputs: IEC 60529 with an IP65 minimum target for the assembled enclosure, IEC 62619 for the battery system, UN 38.3 transport qualification for the assembled pack rather than the bare cells, the IEC 61000-4 series for ESD, EFT, and surge immunity direction, and IPC-class design and workmanship practice together with the fabricator's published DFM limits for the board itself. The radio should be a pre-certified module chosen per deployment country; no frequency band is assumed because southern Africa is not a single regulatory domain.

12. Open questions

Deployment site: the exact country and location set the RF band, the real solar resource, and the enclosure thermal profile.

Radio: the specific modem fixes UART baud, reset polarity, IRQ behavior, true peak current, and cable length; it is also worth asking whether brief power cycling is permitted, since a host-side load switch would cut idle consumption by an order of magnitude if the radio tolerates it.

Enclosure: geometry, material, color, and orientation determine whether +70 C is realistic and how much thermal bias the SHT45 sees.

Battery pack: whether the product uses loose certified cells, an in-house pack, or a third-party pack decides fusing, insulation, NTC attachment, and transport qualification.

Autonomy: if the requirement is several weeks without sun rather than one, battery capacity grows before panel wattage does.

Measurement accuracy: the SHT45 is accurate, but the product-level uncertainty includes enclosure gradients and vent geometry.

BMS telemetry: continuous host communication across the low-side protection ground boundary needs careful treatment, so it matters whether autonomous protection with periodic service access is enough.

Surge: solar and radio cable lengths plus the site grounding arrangement set the formal immunity target.

13. Preliminary major BOM

Qty	Function	Part or requirement
1	Solar charger and power path	BQ25798RQMR
1	Charger inductor	2.2 uH, Isat >= 3 A
1	Battery monitor and protector	BQ76905RGRR
2	Battery protection NFETs	CSD18532Q5B class, TI reference topology
1	Current shunt	10 mohm, >= 1 W, 1%
1	3.3 V buck regulator	TPS62130A
1	MCU	STM32U031K8
1	Environmental sensor	SHT45-AD1F
1	Solar input fuse	1 A one-time
1	Solar connector	2-pin keyed M8 bulkhead
1	Battery connector	4-pin locking
1	Radio connector	8-pin locking
1	Radio bulk capacitor	~220 uF low ESR
1	Charge-enable MOSFET	BSS138 / 2N7002 class
1	Input transient suppressor	TBD after clamp analysis
misc.	Passives	X7R/X5R capacitors, 1% resistors

Check high-value MLCCs around the charger for effective capacitance under DC bias and temperature before the BOM is frozen. Final part selection prioritizes electrical margin, temperature rating, lifecycle, and distributor availability ahead of cost.

14. Scope, validation plan, and tools

I spent the time box on architecture, battery safety, charging and power-path behavior, energy sizing, thermal and environmental risk, and failure modes, and consciously left detailed routing, manufacturing outputs, exhaustive passive selection, enclosure CAD, formal thermal and EMC simulation, and finished firmware outside it. The immediate next steps are schematic and ERC review, layout to the Section 10 floorplan, and prototype bring-up.

The prototype program runs five boards: power bring-up (rails, solar startup, charger configuration, CE behavior, transient probing), firmware and communications, thermal soak at +70 C with charger and radio active, long-duration energy and MPPT logging, and a clean field-reference unit. The validation tests I care most about include every charger startup permutation including solar insertion with battery present and MCU reset mid-charge, per-cell OV and UV, NTC open and short, cell-tap open wire, repeated 1 A transmit bursts with rail droop measured at the radio connector, SHT45 readings against a calibrated probe with the charger active and the enclosure sun-soaked, and forced recovery paths for radio lockup, charger comms loss, and watchdog reset.

I used vendor datasheets, TI reference designs, and an AI assistant during drafting and design review. I cross-checked critical component parameters, charger behavior, and design calculations against manufacturer documentation.

15. Conclusion

The Rev A architecture deliberately separates energy management, battery protection, and supervision. The BQ25798 harvests solar energy and manages normal charging; the BQ76905 protects the cells autonomously, regardless of what the charger or firmware believes; the STM32 configures, monitors, and recovers. The 10 W panel exists for environmental margin rather than steady-state demand, the pack supports roughly a week without sun at the chosen 5-minute cadence, charging is inhibited whenever pack conditions are unsafe, and every fault path degrades functionality before it touches battery safety. Rev A favors safe failure behavior, thermal margin, and diagnosability over aggressive cost or area optimization, which is the right trade for the first revision of a product nobody will visit for a year.